

• ADFORGE AI — PROJECT DOCUMENTATION

• AdForge AI - Automated Multi-Platform Copy System

- **Date:** 7th July 2026
- **Development Profile:** Sunehri Sharma
- **Maximum Marks:** 5 Marks
- **Phase:** Project Initialization and Planning Phase

Proposed Overview & Solution Text

- **Proposal Introduction:**

The proposal report aims to transform digital marketing operations using Generative AI, boosting copywriting efficiency, platform compliance, and workflow execution. It tackles system inefficiencies and manual text formatting bottlenecks, promising faster deployment cycles, reduced campaign spend risks, and a superior creation experience. Key features include a LLaMA 3.3-powered credit/copy model and real-time asynchronous generation pipelines.

- **Project Overview:**

- **Objective:** The primary objective is to revolutionize the ad copywriting process by implementing advanced large language models (LLaMA 3.3-70B via Groq API), ensuring faster, tone-adaptive, and highly accurate marketing copy variations.
- **Scope:** The project comprehensively assesses and enhances the ad variant setup workflow, incorporating a lightweight local Python environment to render, refine, and export compliant multi-channel ad formats.

- **Problem Statement:**

- **Description:** Addressing formatting inaccuracies, manual re-writing fatigue, and character constraint complexities in current multi-platform marketing workflows that adversely affect a business owner's operational efficiency and time-to-market.
- **Impact:** Solving these issues will result in dramatically improved creation speed, eliminated visual constraint errors (e.g., Google's 30-character limit violation), and optimized target messaging, contributing directly to higher click-through rates (CTR) and overall campaign scaling success.

- **Proposed Solution:**

- **Approach:** Employing cutting-edge inference techniques via the Groq Cloud API to programmatically parse marketing briefs and dynamically adjust copywriting structures to match network-specific UI requirements.

- **Key Features:**
 1. Micro-workspace interface delivering synchronous variations across Google, Facebook, and Instagram instantly.
 2. Enforced structural algorithms guaranteeing character-limit compliance for Google text search elements.
 3. Asynchronous interactive "Refine Copy" feedback workflow allowing instant modifications based on creative prompts.

Proposed Resource Requirements Allocation

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU specifications and runtime execution architecture	Local Core Processing Unit (Intel i5/i7 or Apple Silicon M-Series, min. 4 cores)
Memory	RAM specifications for running local web scripts	8 GB or 16 GB RAM
Storage	Disk space for workspace environment files, local scripts, and dependency libraries	256 GB or higher SSD
Software		
Frameworks	Python-based web application web layer framework	Flask
Libraries	Additional backend integration and environment libraries	groq, python-dotenv, fpdf2 (or standard formatting engines)
Development	Local Integrated	Visual Studio Code (VS Code), local modern

Resource Type	Description	Specification / Allocation
Environment	Development Environment (IDE)	web browser
Data		
Data	Input payload scope, formatting schemas, and source types	Dynamically supplied text arrays (Product details, brand goals, target channels, selected tone attributes). Zero data persistence for enhanced user privacy.